

Paper	ENCN342 – Fluid Mechanics and Hydraulics
Topic	Steady Open Channel Flow
Module	1 – Basic Concepts

Summary

This module introduces the topic of open channel flow. The key equations are derived and the steady open channel flow problem contrasted with the steady pipe flow problem.

Prerequisites

You should be familiar with control volume analysis techniques, and the associated ideas of conservation of momentum, mass and energy.

Objectives

At the conclusion of this module you should:

1. Appreciate the sort of questions an engineer may wish to answer regarding open channel flows.
2. Be able to derive the governing equations for conservation of mass, energy and momentum, and be aware of the assumptions made in their derivation.
3. Be able to explain the differences between the transition problem for channel and pipe flow.

References

Hunt 1995 pp 4.1-4.26.

Key Points

1. Open channel flow

Open channel flows are free-surface flows. An engineer has interest in these flows for reasons ranging from channel management, to the disposal of effluent in the channel.

2. Governing equations

The analysis of open channel flow problems will be done using control volume techniques. The three governing equations are

Conservation of mass

$$u_1 A_1 = q_1 B_1 = u_2 A_2 = q_2 B_2 = Q = \text{discharge} = \text{constant} \quad (1)$$

where u is the cross-sectionally averaged velocity, A is the cross-sectional area of the channel, B is the channel width, and q is the **discharge per unit width** (m^2/s) defined by

$$q = uy = \frac{Q}{B} \quad (2)$$

where y is the flow depth.

Conservation of energy

$$y + z + \frac{u^2}{2g} = H \quad (3)$$

where z is the height of the channel bed. In deriving equation (3) it has been assumed that the channel bed and free surface have a very small slope. This assumption has allowed us to take the pressure distribution in the channel to be **hydrostatic** in the vertical. It has also been assumed that, H , the total energy is constant. In other words energy losses are negligible. In applying equation (3) these assumptions must not be forgotten.

Conservation of momentum

$$(q\rho u)_2 - (q\rho u)_1 = \frac{\rho g y_1^2}{2} - \frac{\rho g y_2^2}{2} - p_f \quad (4)$$

where p_f is any other force (per unit width) acting along the channel, taken to be positive when acting upstream.

3. The transition problem

The principal difference between pipe flow and channel flow calculations is illustrated in the transition problem where the pipe cross-section changes, or the channel bed elevation changes. In pipe flow the velocity after the transition can be found from the conservation of mass equation. In channel flow, because the height of the free surface is unconstrained, the velocity and flow depth downstream must be calculated simultaneously. This requires the use of the conservation of mass and energy equations (assuming that there are no energy losses). A simple rearrangement of the energy equation demonstrates that there can be more than one solution for the flow depth and velocity downstream.

Worked Examples

Question 1

A sluice gate is placed in a rectangular channel that is 8 metres wide, and that carries a discharge of 16 cumecs ($16 \text{ m}^3/\text{s}$). The upstream flow depth is 2.52m and the downstream flow depth is 0.30m .

- (a) Confirm that these flow parameters correspond to a negligible energy loss through the gate.
- (b) Calculate the force on the sluice gate.

Solution

- (a) The cross-sectional average velocities can be found from the conservation of mass equation.

$$u_1 = \frac{Q}{By_1} = \frac{16}{8 \times 2.52} = 0.79 \text{ m/s} \quad (1)$$

and

$$u_2 = \frac{Q}{By_2} = \frac{16}{8 \times 0.30} = 6.67 \text{ m/s} \quad (2)$$

The total energy in each cross-section (we will assume that the bed elevation does not change through the gate, and hence that the bed elevation is 0) is

$$\boxed{y_1 + z_1 + \frac{u_1^2}{2g} = 2.52 + \frac{0.79^2}{2 \times 9.81} = 2.55 \text{ m}} \quad (3)$$

$$y_2 + z_2 + \frac{u_2^2}{2g} = 0.30 + \frac{6.67^2}{2 \times 9.81} = 2.57 \text{ m} \quad (4)$$

Thus the energy losses through the gate are negligible.

- (b) The force on the gate can be calculated from the momentum equation

$$(q\rho u)_2 - (q\rho u)_1 = \frac{\rho g y_1^2}{2} - \frac{\rho g y_2^2}{2} - p_f$$

The discharge per unit width, q , is $2 \text{ m}^2/\text{s}$. So we have

$$\begin{aligned}
 p_f &= \frac{\rho g y_1^2}{2} - \frac{\rho g y_2^2}{2} - (q\rho u)_2 + (q\rho u)_1 \\
 &= \rho \left(\frac{9.8 \times 2.52^2}{2} - \frac{9.8 \times 0.3^2}{2} - 2 \times 6.67 + 2 \times 0.79 \right) \quad (5) \\
 &= 1000 \times 18.95 \\
 &= 18.95 \text{ kN / m}
 \end{aligned}$$

This is the force per unit width, so the total force is obtained by multiplying by the channel width

$$F_{gate} = 151.6 \text{ kN} \quad (6)$$